Abdullahi Mohamed

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EDUCATION

The Ohio State University Columbus, OH
Bachelor of Science in Computer Science, Minor in Applied Mathematics, GPA: 3.8 Expected May 2028
Relevant Coursework: Data Structures and Algorithms, Operating Systems, Distributed Systems, Database Systems, Computer Networking, Computer Architecture, Applied Machine Learning, Discrete Structures

TECHNICAL SKILLS

Languages & Databases: Python, Java, Go, C#, JavaScript, TypeScript, C/C++, MySQL, PostgreSQL, DynamoDB, CosmosDB
Frameworks & Libraries: Spring Boot, React, FastAPI, Flask, .NET Core, PyTorch, LangChain, Pandas, vLLM, TensorRT
Cloud & Infrastructure: Docker, Kubernetes, Azure (AKS, Functions), AWS (Lambda, EKS, S3), CUDA, Kafka, Redis
Developer Tools: Git, Linux, Gradle, Maven, Jenkins, Azure DevOps, GitHub Actions, JUnit, Postman

WORK EXPERIENCE

Intuit Mountain View, CA
Software Engineer Intern May 2026 – Aug 2026

- Reduced manual incident triaging by **90%** through building an agentic AI response platform using LangGraph and FastAPI on Kubernetes, automating diagnostics and state-based routing across **25+** production microservices
- Saved **200+** engineering hours weekly by orchestrating automated workflows across PagerDuty, Splunk logs, and SQL query execution profiling to securely process **1,000+** infrastructure alert investigations per week
- Enabled autonomous resolution for **75+** weekly incidents by engineering a Retrieval-Augmented Generation (RAG) pipeline using semantic chunking and a vector database to index and query **50+** internal service runbooks
- Eliminated **80%** of manual configuration for onboarding new services by building a developer self-service portal, enabling rapid service registration and automated API key generation across internal engineering teams

The Ohio State University Columbus, OH
Software Engineer Research Assistant Aug 2025 – Apr 2026

- Engineered a concurrent Python multiprocessing pipeline to extract and normalize **120,000+** DICOM imaging slices, optimizing disk I/O bottlenecks and standardizing clinical metadata for advanced digital twin modeling
- Boosted vertebral fracture prediction accuracy by **15%** over published baseline models by building a PyTorch benchmarking framework to evaluate and hyperparameter-tune 3D CNN and Vision Transformer architectures
- Decreased model training time by **40%** by containerizing a distributed PyTorch environment with Docker and scaling execution across lab GPU clusters to eliminate dependency bottlenecks

PROJECTS

ServerFlow | Distributed LLM Inference Platform (Go, Kubernetes, vLLM, Redis, Kafka)

- Engineered a distributed Go LLM inference router on Kubernetes, load-balancing requests across autoscaled vLLM workers based on model availability, real-time GPV-cache load, and queue depth
- Implemented zero-downtime failover via Redis-backed state coordination and Kafka event streaming, enabling automated worker health checks, request rerouting, and real-time latency telemetry

MotionMan | Object Tracking Engine (C++, CUDA, TensorRT, RT-DETR, OpenCV)

- Architected an asynchronous, multi-threaded C++ video pipeline using RT-DETR and ByteTrack, sustaining 40+ FPS tracking across **25+** streams while reducing identity switches by **35%** under severe occlusion
- Accelerated end-to-end throughput by **1.8x** and slashed inference latency by **45%** by compiling FP16 TensorRT execution engines and optimizing host-to-device CUDA memory transfers

BuckeyePlanner | OSU Degree Audit Platform (Java, Spring Boot, PostgreSQL, Docker, AWS EKS)

- Developed a scalable Java/Spring Boot degree audit engine for **60,000+** students, automating prerequisite validation across **200+** programs and boosting course recommendation accuracy by **25%** via graph traversal
- Containerized backend microservices and PostgreSQL databases using Docker on AWS Kubernetes (EKS), establishing CI/CD pipelines that streamlined infrastructure and reduced deployment times by over **80%**

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Minorities in Technology at The Ohio State University Columbus, OH
Chapter Founder & President Aug 2025 – Present

- Increased member engagement by **40%** by spearheading chapter operations and weekly technical workshops
- Secured strategic recruitment partnerships with Microsoft, Google, and LinkedIn to accelerate career placement